How Time-MoE makes a forecast

Shi et al. · 113M parameters, 50M used per step (TimeMoE-50M) · decoder-only mixture of experts · zero-shot

1. One series at a time

Time-MoE reads only the target's own past. A ? is a value not known yet.

past *future*

Target temperature

No covariates; each metal on its own.

Context plus horizon must stay within 4,096 steps.

2. Every step a token

The model does not scale its input, so our code scales the context by its mean μ and standard deviation σ first.

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

one value per token

gated layer

8 tokens of 384 numbers

3. Twelve causal blocks with experts

Each token reads itself and earlier tokens. RMSNorm in front, shortcut around.

earlier → later

earlier steps only

Self-attention: causal, 12 heads, rotary positions (RoPE)

gate

2 of 8 experts, plus a shared one

Experts: a softmax gate picks 2 of 8 SwiGLU experts (768); a shared expert (1,536) always runs, weighted by a sigmoid gate

4. Heads for 1, 8, 32 or 64 steps

Each pass uses the **longest head that fits** the steps still missing, appends the result and runs again.

point forecast

90 days = 64 + 8 + 8 + 8 + 1 + 1: six passes. A point forecast only.

Sparse experts give a large model at the cost of a small one, with every step read as its own token.

12 blocks · width 384 · 12 heads · 8 experts, 2 per token, plus 1 shared · output heads of 1, 8, 32, 64

In the notebooks: last 2,048 days or full monthly history · each metal on its own · no covariates · no intervals